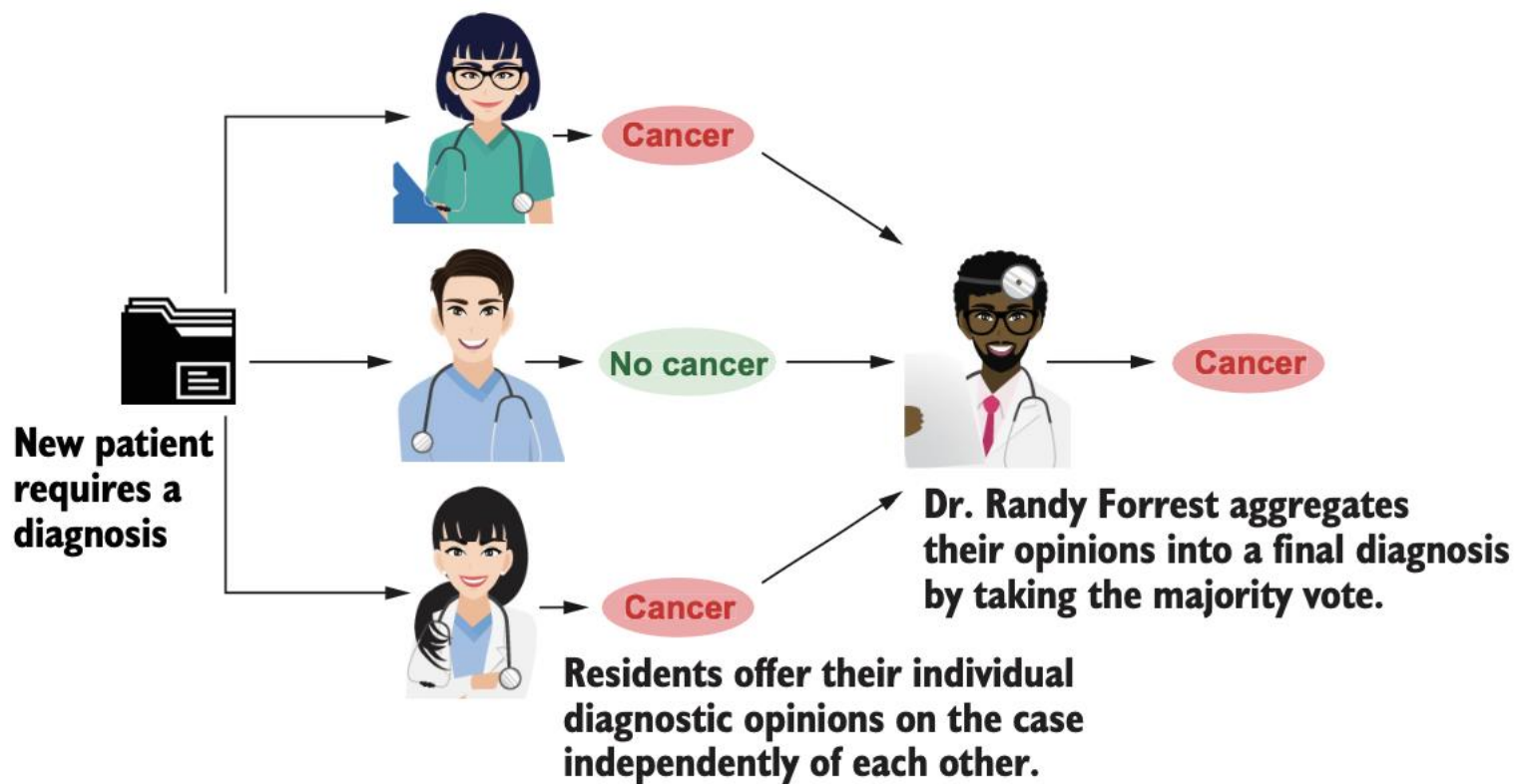


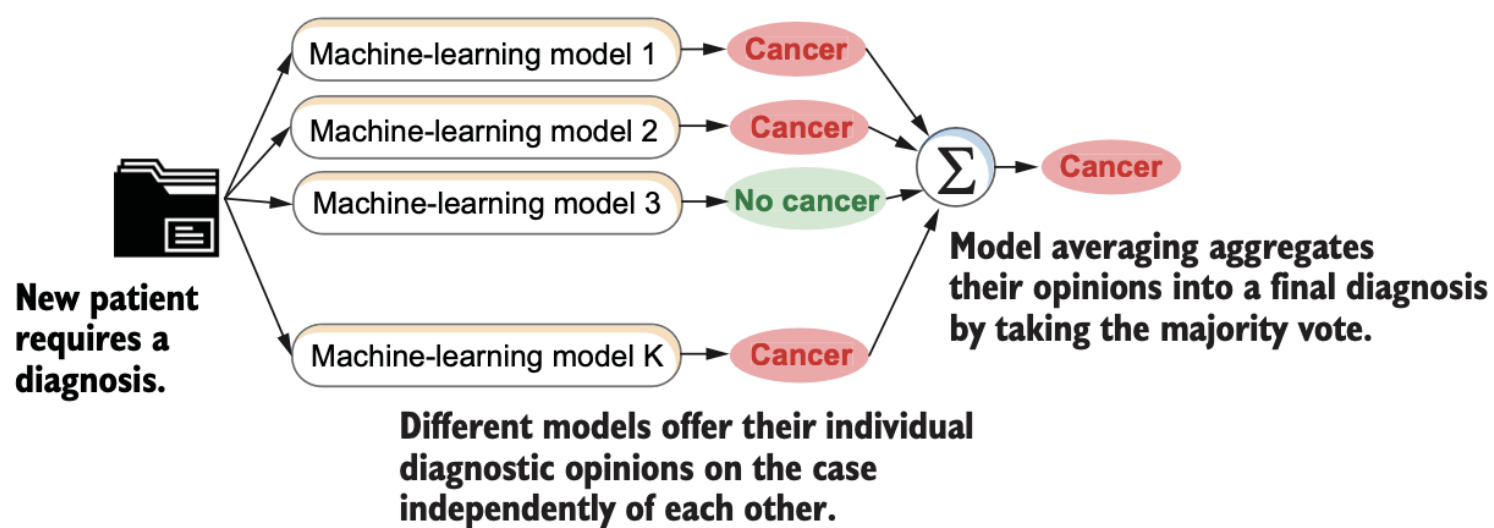
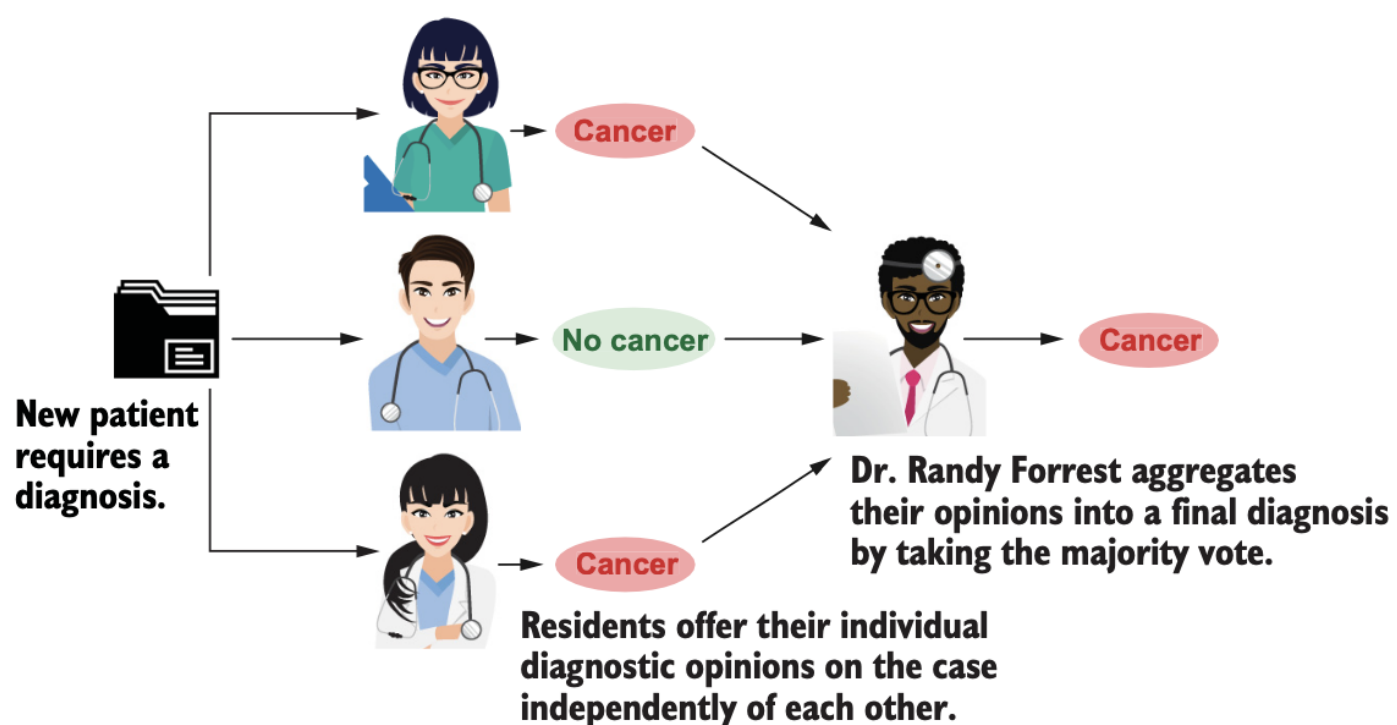
Bagging & ensemble modeling

2026.04.07

Ensemble Models



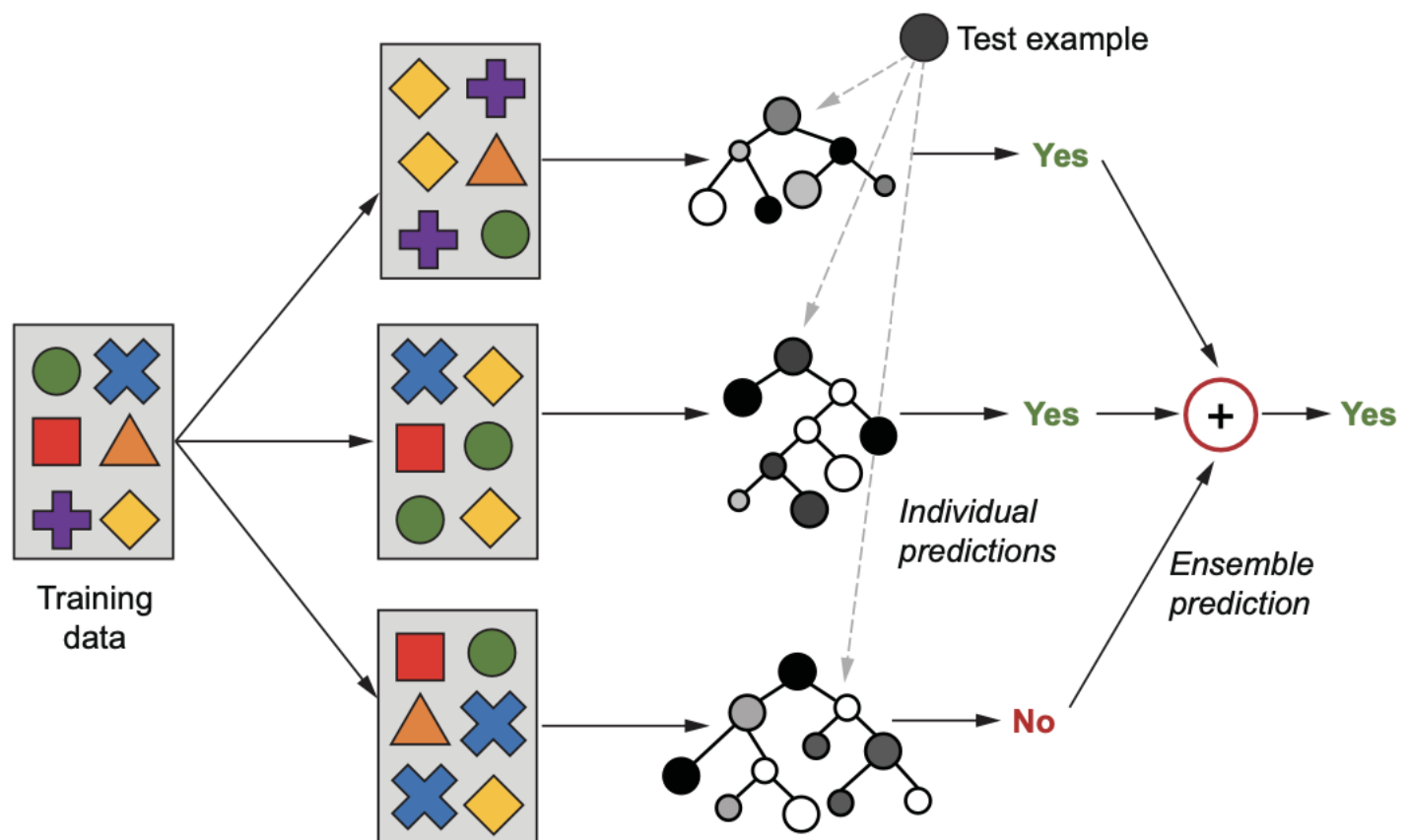
Ensemble Models



Ensemble Models

1. Bagging
2. Boosting
3. Stacking

Bagging



Bootstrap sampling
generates diverse subsets
for training base learners.

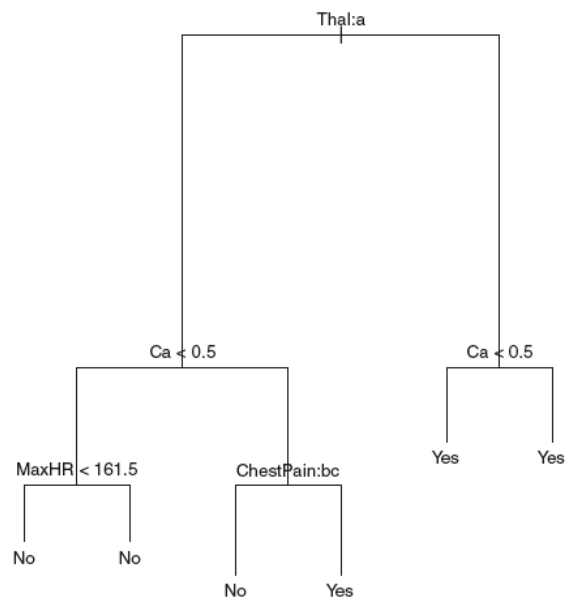
Diverse **base learners**
are trained on sampled
subsets of the data.

Final prediction of
the ensemble is reached
by **model aggregation**.

Bagging

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

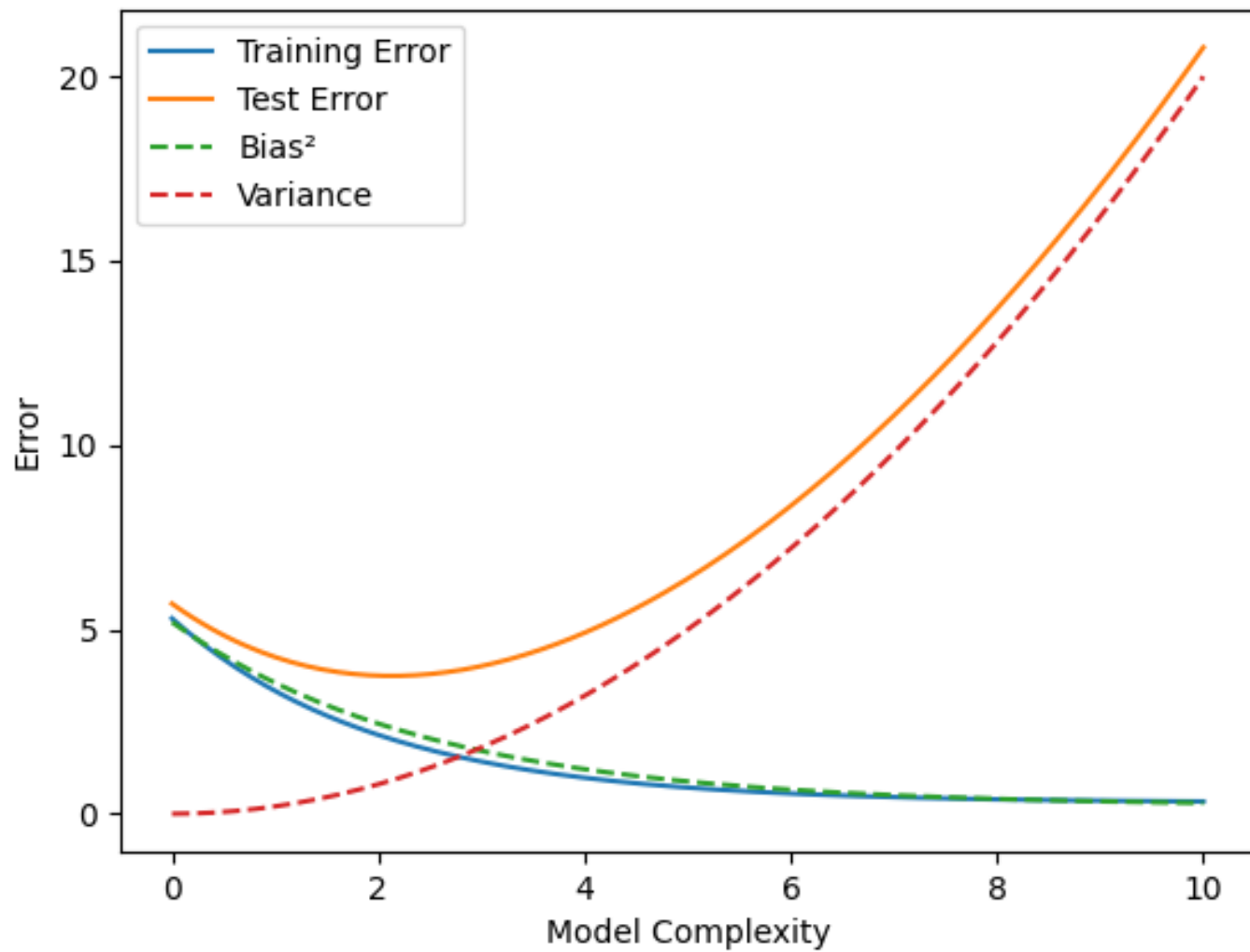
Decision Trees



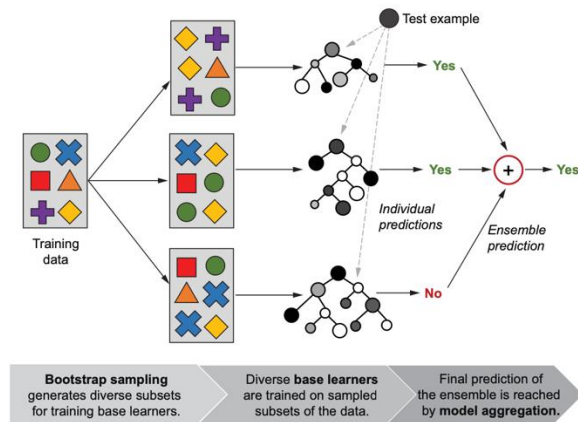
Advantages:

Disadvantages:

Bias vs variance



Bagging Tree Ensembles



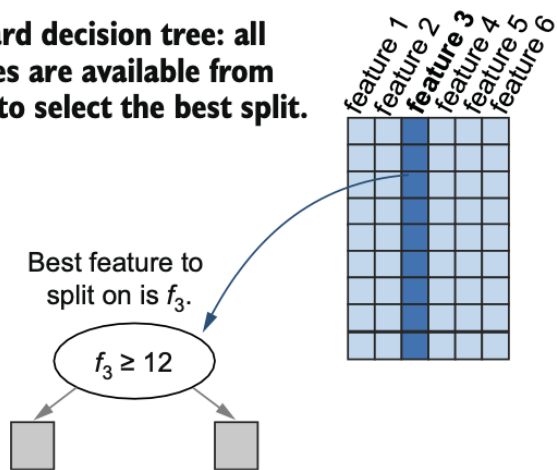
Each tree uses all features in its bagged dataset at each split in the tree

BaggingClassifier *can use default DecisionTree or other base model

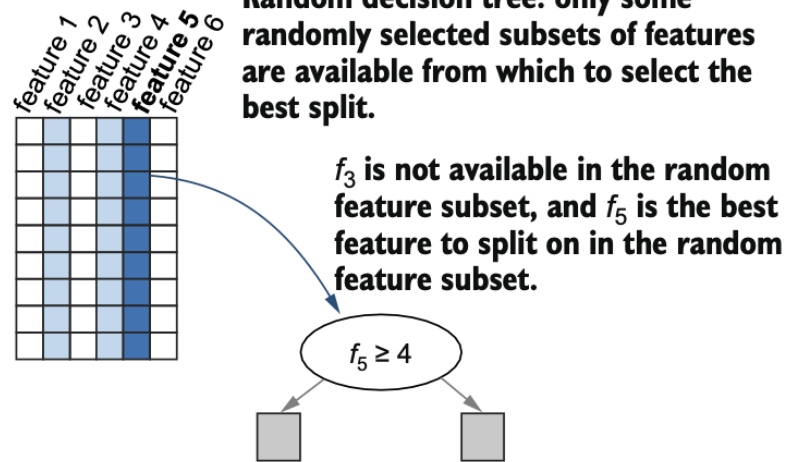
BaggingRegressor *can use default DecisionTree or other base model

Random Forest Models

Standard decision tree: all features are available from which to select the best split.



Random decision tree: only some randomly selected subsets of features are available from which to select the best split.



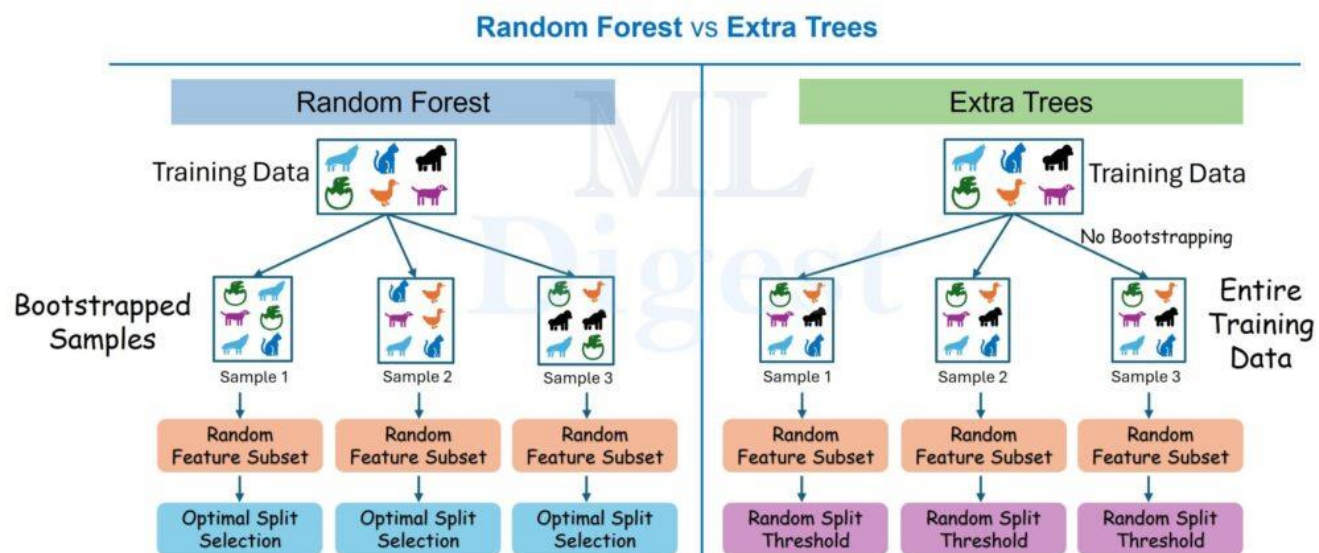
Random Forest Models

- Strengths
- Weaknesses

[RandomForestClassifier](#)

[RandomForestRegressor](#)

Extra Random Tree Models



<https://ml-digest.com/wp-content/uploads/2025/12/random-forest-vs-extra-trees-2048x1177.jpg>

ExtraTree Models

- Strengths
- Weaknesses

[ExtraTreesClassifier](#)

[ExtraTreesRegressor](#)

Random Forest Parameters

RandomForestClassifier

class

```
sklearn.ensemble.RandomForestClassifier(n_estimators=100,  
*, criterion='gini', max_depth=None, min_samples_split=2,  
min_samples_leaf=1, min_weight_fraction_leaf=0.0,  
max_features='sqrt', max_leaf_nodes=None,  
min_impurity_decrease=0.0, bootstrap=True,  
oob_score=False, n_jobs=None, random_state=None,  
verbose=0, warm_start=False, class_weight=None,  
ccp_alpha=0.0, max_samples=None, monotonic_cst=None) #
```